



KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching agents to autonomously decide when to act, reflect, or seek knowledge

arxiv.org/abs/2504.03553



The Problem: Indiscriminate "Flood Irrigation" in Agent Training

Trajectory Imitation

e.g., ReAct

Blindly imitates gold trajectories without assessing situational needs

Trial-and-Error

e.g., Reflexion

Applies external feedback uniformly, even when unnecessary

Knowledge-Augmented

e.g., KnowAgent, WKM

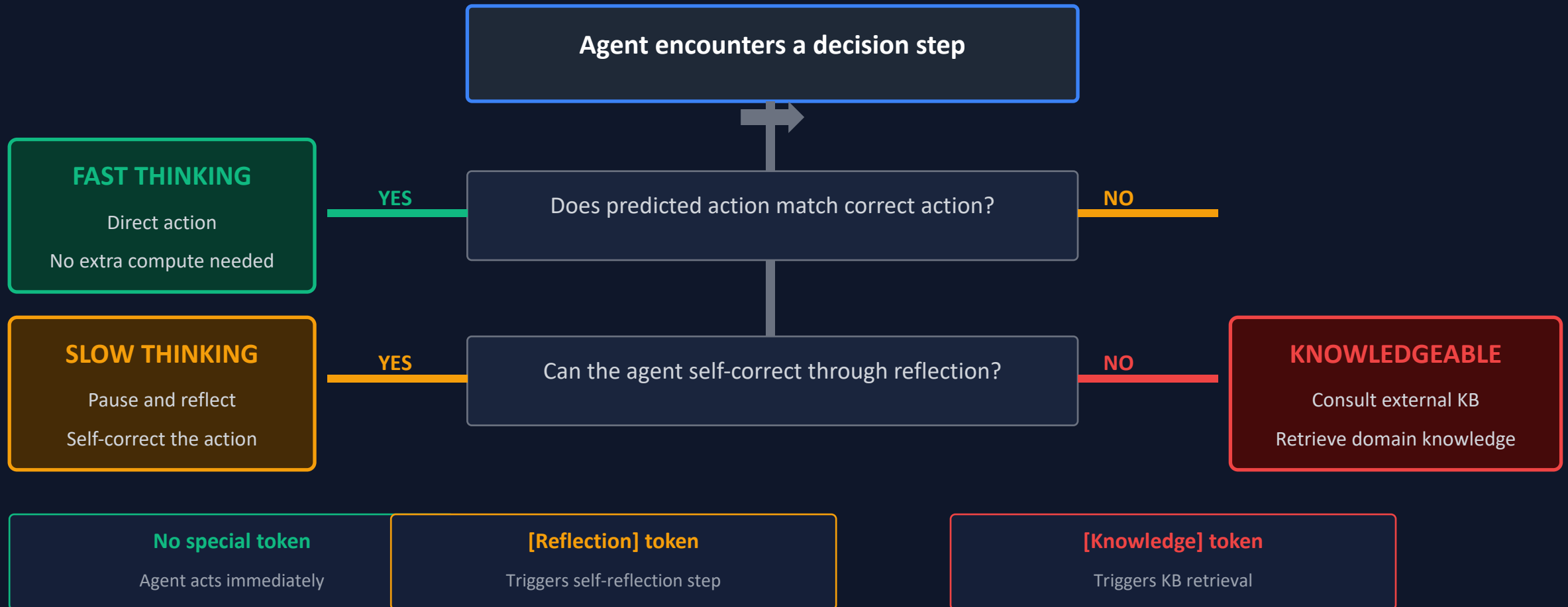
Injects domain knowledge at every step regardless of agent capability

The Core Insight

Humans possess situational self-awareness: the metacognitive ability to assess whether they can handle a situation directly, need to pause and reflect, or must seek external help.

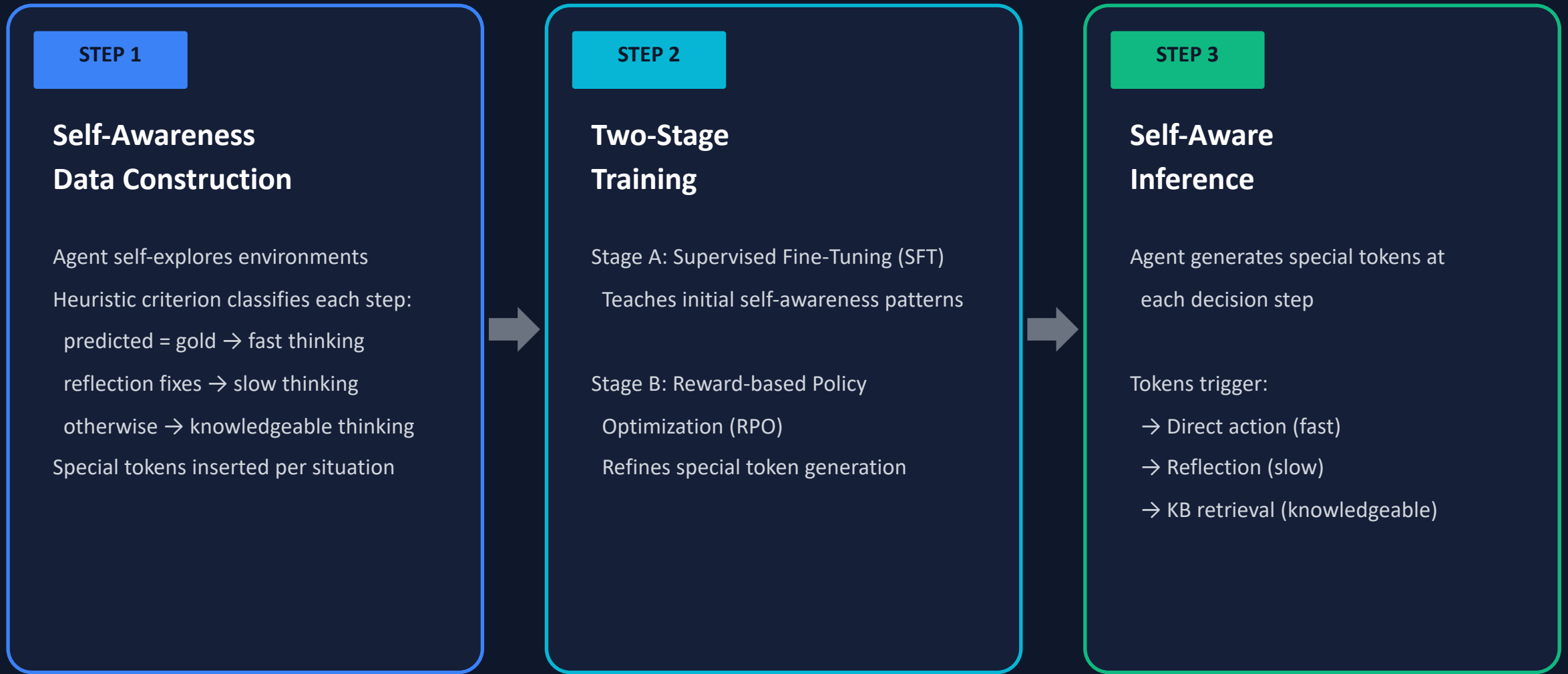
KnowSelf brings this capability to LLM agents — selective, not saturated.

Three Situational Thinking Modes



KnowSelf uses heuristic situation judgment to classify each step and insert the appropriate token during training.

KnowSelf Framework: Three-Stage Pipeline



Lightweight and data-efficient: requires only heuristic labeling on self-explored trajectories, not expensive human annotation.

Qualitative Examples: Where Baselines Fail, KnowSelf Succeeds

Task: "Put a hot mug in cabinet"

OpenAI-O1

FAILS

After opening the microwave, incorrectly tries to take an egg from it instead of heating the mug.

KnowSelf (Llama-8B)

SUCCEEDS

Uses [Reflection] token to realize it should heat the mug using the open microwave.
Slow thinking → self-correction.

Task: "Put a saltshaker in drawer"

DeepSeek-R1

FAILS

Reflects that the drawer is open and tries to close it first — but this action fails and the agent gets stuck.

KnowSelf (Llama-8B)

SUCCEEDS

Uses [Knowledge] token to retrieve guidance: place the object without removing unrelated items. Knowledgeable thinking → correct action.

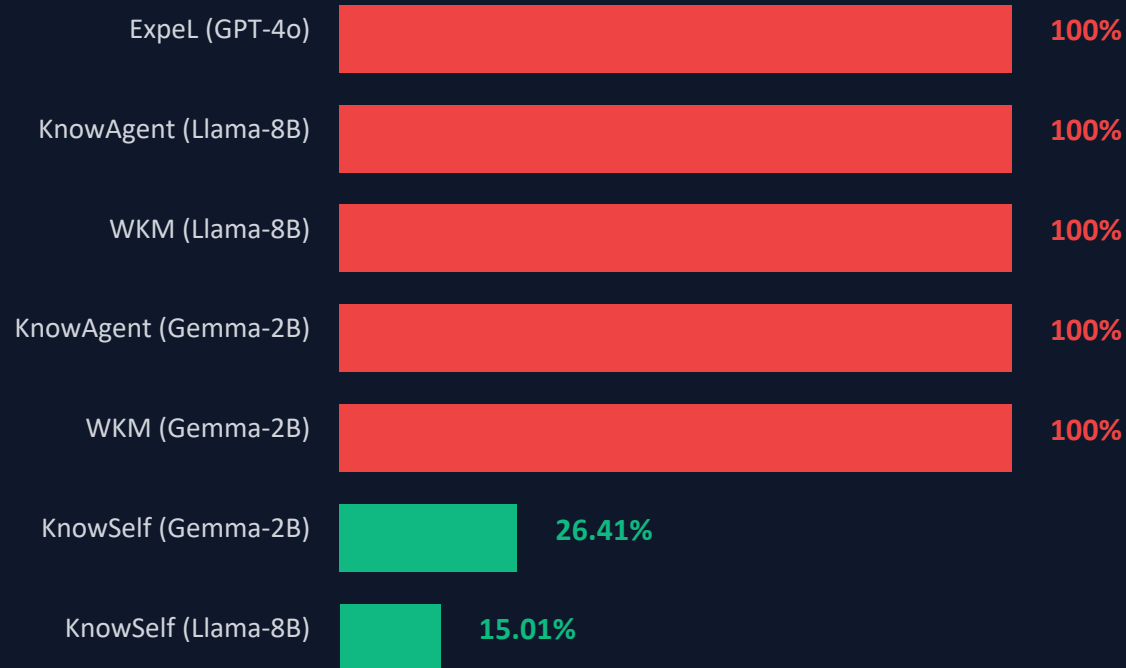
ALFWorld Results: KnowSelf Achieves 84.33% with Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

KnowSelf (Llama-8B) surpasses GPT-4o ExpeL (79.85%) with 84.33% overall accuracy — using only 15% knowledge vs. 100%.

Minimal Knowledge, Maximum Impact: Efficiency Advantage

Knowledge Usage (% of steps using external KB)



Overall Accuracy (%) on ALFWorld



Key Insight: Less Knowledge → Better Performance

- KnowSelf (Llama-8B): 15% knowledge → 84.33% accuracy | Full-knowledge methods: 100% knowledge → 73–80% accuracy
- Even Gemma-2B with KnowSelf (26% knowledge) matches GPT-4o ExpeL (100% knowledge) at 79.85%

Key Takeaways & Future Directions

1

Indiscriminate injection is wasteful

"Flood irrigation" of knowledge hurts agent performance and adds unnecessary inference cost.

2

Agents can learn situational self-awareness

Special tokens with lightweight heuristic labeling enable agents to regulate cognitive resources autonomously.

3

KnowSelf outperforms strong baselines

Llama-8B with KnowSelf: 84.33% vs. GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%) on ALFWorld.

4

Selective > saturated knowledge use

Only 15–26% knowledge usage needed versus 100% for ExpeL, KnowAgent, and WKM.

5

Scales across model sizes

Even Gemma-2B reaches 79.85% with KnowSelf, matching GPT-4o ExpeL — the paradigm generalizes downward.

Future: Multi-domain generalization · Dynamic knowledge base expansion · Integration with reasoning-specialized models